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When a Design Method Changes: Re-execution and Improver Lineages in Battery-Free Design

Working draft

Complete experimental manuscript, 20 October 2026

Abstract

A successful device does not by itself validate the method that designed it, and a new method does not establish an improved method proposer. We study these distinctions in a fixed executable battery-free apparatus. Our protocol separates device programs D, design procedures H, and procedures G that generate H; it requires fresh decisions after a global method change while allowing input-identical lower-level response reuse. An initially recovered model-authored method passes its device checks but fails independent promotion because initial cache parity is missing and its proposed necessary constraints exclude a passing baseline. A prospective comparison then yields 12/12 service passes for each of fixed H, ordinary reflection, and a typed procedure, using 64, 117, and 127 requests respectively. In a subsequent frozen transfer study, selected descendants yield 7/9, 6/9, and 8/9 passes for H_0, G_0, and G_1; four token-censored attempts remain in the denominators. The three lineage pairs share one G_1 formation, so these differences are descriptive. All 1,238 finite-campaign requests cost USD 10.0268025 at the frozen upper rate estimate. We retain the unchanged base method, provide saved-data reproduction, and preserve the A/R product's dark-service and FIFO failures. No stable recursive improvement or physical qualification is established.

1 Introduction

An agent designing an energy-harvesting device may change a capacitor, rewrite firmware, revise its experiment order, or change the instructions used to propose future revisions. These actions intervene on different objects. Checking a new capacitor under old recorded controller actions can give the wrong physical response. Checking an old device after rewriting the design prompt does not measure the new prompt's design decisions. Selecting a useful prompt does not show that the procedure which generated it has improved. A long history of tool calls makes these distinctions harder to audit unless the experiment records both the intervention and the evidence it invalidates.

Battery-free design makes the problem concrete. Harvested input, storage voltage, boot timing, event capture, transmission, and later controller actions affect one another. A reset after an event was delivered can be harmless for a single-event task, while a reset with pending volatile state can destroy service. For a history tag, retaining already sampled records does not establish continued sensing in darkness. An apparently successful upload can also conceal delayed fresh samples or a violation of the required queue order. These distinctions require the actual service and temporal record, rather than an agent's verbal diagnosis.

We use a fixed executable scientific apparatus to study device-level responses, fresh method execution, and

method-generating procedures. The apparatus and external evaluator are held outside the participant's authority. Tools expose source-bound facts; the participant can propose a legal artifact but cannot declare acceptance or edit the service to make its artifact pass. We retain rejected calls, resource stops, unused siblings, posthoc adaptations, and unresolved accounting alongside favorable results. The resulting experiment can reject an update without treating rejection as a failure of the research process.

The empirical story has two successive tests of stronger claims. First, a method recovered from a failed live proposal produces a successful device, but the independent confirmation gate rejects promotion. A later prospective comparison matches initial caches and includes a strong fixed-method control. All three method conditions pass every formal task/repetition; the two proposal conditions consume more requests. Second, we compare ordinary reflection G_0 with one actual model-authored proposer G_1, using paired descendant lineages and frozen future inputs. The final counts differ, but the two digital derivatives saturate every arm and the remaining service task contains both submitted failures and token censoring. These observations do not establish a stable improvement process.

Our contribution is a source-bound experimental separation and its measured limits: a typed validity contract for $D/H/G$ changes; a complete negative method comparison after a failed initial confirmation; a conditional improver-lineage study with all formation, sibling, selection, failure, and confirmation costs; and a portable readback of the saved evidence. We do not claim a new general incremental simulator, superiority to a state-of-the-art prompt optimizer, physical device acceptance, or an autonomous research system without developer intervention.

2 Experimental objects and validity contract

Let W denote the fixed world, T the task and service contract, D the device program, and E the external evaluator. The complete response $Y = F(W, T, D)$ contains the coupled physical and controller trajectory, and $E(T, D, Y)$ produces the computational service result. A design procedure H chooses observations, orders experiments, and produces D from a fresh task entry. A proposer G reads an admitted development history and generates candidate H artifacts for a registered selection procedure. The device's runtime, an agent's decision span, and successive method generations are different experimental axes.

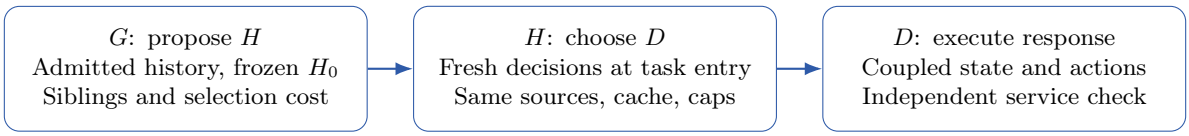


Figure 1: Three intervention objects. Task, world, evaluator, and budget authority remain external. An arrow transfers a frozen artifact; it does not transfer an acceptance verdict. Exact-input response reuse is separate from fresh decisions.

For a changed D, a reusable record must match all consumed inputs, apparatus identities, numerical settings, and output bindings. The old trajectory is an explicit negative control when a coupled input changes. Dependency and boundary/scope invalidation may conservatively expand to the complete feedback response. An exact-input cache can reuse a qualified response; building that cache still has a cost. An evaluator-only change may reuse the same raw response while requiring a new verdict. It cannot silently replace the previous verdict.

For a global $H \to H'$ change, our conservative policy restarts decisions at task entry with a new owner and empty decision context. It does not replay the old agent's reasoning as evidence for H'. Qualified lower-level responses may still be reused when their consumed inputs match. A smaller restored decision suffix would need its own state-validity proof; no such advantage is reported here. A G comparison additionally fixes H_0, the allowed development history, selection tasks, future-input exclusion, and total improvement budget. Every candidate branch, including an unselected sibling, remains part of that cost.

Promotion is a separate decision from schema eligibility and service success. The method artifact records a hypothesis, falsifier, typed change and scope. The external registry checks source, tool, model configuration,

evaluator, initial cache, fresh-context and cost identities. A structural pass admits an artifact for testing; it does not validate the artifact's scientific claims. The default method changes only under the registered retention rule.

3 Apparatus, tasks, and measurements

The principal apparatus is an ideal digital battery-free model with piecewise source input, finite storage, conversion and sleep costs, boot and loaded-voltage thresholds, event capture, and a receiving endpoint. In the small BF tasks, legal device edits are capacitance and the earliest allowed transmit time. Event, source, initial state, all other device fields and evaluator stay fixed. The service requires the specified application frame by its deadline without duplicate acceptance. Independent consumers check energy, receiver behavior, transient voltage and deduplication. Passing these computations is not physical calibration or an official benchmark result.

The prospective method comparison uses four fixed task IDs: fast-pulse BF, volatile-reset BF, latched-reset BF, and a synthetic C-Lite single-query DC derivative. They exercise distinct mechanisms within a small declared model, not four independent hardware families. The C-Lite case does not qualify RF, nonce security, multiple-query operation, or a real radio deployment. The formal task list is finite; repeated sampling on a task is not an additional task.

The continuous A/R case concerns an indoor-light-powered temperature/history tag. It couples photovoltaic port power, a capacitor, regulator thresholds, sensing, persistent writes, record ranking, transmission and a digital gateway. Source power is specified in watts; illuminance is descriptive metadata. The legal apparatus starts with no stored energy and has an uncalibrated source, storage and write model. It provides an engineering counterexample to broad completion claims, not a physical demonstration. Section 7 preserves that case's separate service and ordering requirements.

We measure independently evaluated service, participant service judgment, ordered capture–reset–delivery events, admitted response identities, provider requests, token-based charge estimates, summed episode wall time, failed actions, resource censoring, native dispatch and new scientific execution. Requests, logical design checks, cached response files and physical experiments are not interchangeable counts. Public feedback is available to the participant; the external final verdict remains separate. Developer assistance, model-weight identity and physical qualification have their own boundaries.

4 From recorded failures to an independently rejected method

4.1 Response controls and an initial prompt probe

Six deterministic public controls first compare frozen-trajectory replay, dependency invalidation, boundary invalidation, full response execution and exact-input response caching. The apparatus has arrivals, finite capacity, a threshold controller and action-dependent consumption. The unchanged case is a positive reuse control. Changing energy, load, threshold, capacity or another consumed input makes the frozen trajectory disagree with the full response in the five changed cases. Dependency, boundary and exact-input strategies match full response on all six. In coupled cases the two invalidation strategies expand to full response; this is a correctness qualification with no demonstrated physical-solver speedup.

An initial public firmware task then tests actual method re-execution, rather than replay. The power board is fixed and the participant revises firmware to deliver six valid frames in twenty seconds. The operator-authored H_1 adds a constraint and time-origin audit to the base instruction. Four fresh attempts run in order H_0, H_1, H_1, H_0 with the same initial artifacts and native-cache boundary. Each permits at most 80 parent-plus-child requests, four candidate evaluations, and 1,200 seconds. Both H_1 attempts and the first H_0 attempt deliver qualified computational designs. The final H_0 attempt stops at the frozen disk-headroom gate before validation. It remains a censored attempt, not a replacement opportunity. All 68 requests settle, at USD 1.1429220 upper. The prompt was supplied by the operator, not an autonomous G, and the default method remains unpromoted.

4.2 A real model-authored method, recovered posthoc

In the next method episode, the model is asked to propose a repeatable H. The live proposal instead encounters a wrongly exposed host action and a candidate-slot contract failure: a device-check payload consumes a method slot. The episode uses nine requests and ends incomplete with no successful method submission. A posthoc host adapter recovers the actual model-authored method from the retained rejected call, adding no paid proposal or new model decision. The recovered artifact is eligible for experimental re-execution; the original live episode remains a failure.

The proposed method acquires sources, constructs an energy/timing ledger, tracks volatile event loss, and checks a design. It also states task-specific exclusions: a necessary 7.4–14 μF band, rejection of waiting across the dark gap or using a late transmit floor, and a recommended 10 μF / 2 ms candidate. These statements are scientific claims, not consequences of a valid method schema. The independent test must retain both the proposed falsifier and any over-restrictive rule.

Fresh H_0 and H' entries each begin with zero decision messages, distinct owners, the same task and sources, and caps of two candidates, 24 requests and 1,200 seconds. Each checks and submits one candidate. The base method chooses 22 μF / 45 ms; the revised method chooses 9 μF / 3.7 ms, rather than copying the recommended values. Both pass the full frozen computational service. First reception is 46.9 ms and 5.6 ms respectively. These are fresh device decisions; external readback reuses their exact-input complete responses.

4.3 Independent confirmation rejects promotion

Two observations prevent a method-benefit claim. First, the physical cache evolved between entries and its initial snapshots were not frozen equally. The registry explicitly rejects initial-cache parity. Input-correct reuse of an individual response does not establish equal starting cache exposure. Second, the successful 22 μF / 45 ms baseline contradicts the proposed necessary band and rejection of a late floor. It does not test the separate specific 22 μF / 2 ms edge case or the recommended device, and does not refute the distinct claim that transmission fits within one contiguous segment. A successful revised device cannot validate every rule of its method.

The event-order audit also corrects a tempting mechanism interpretation. The baseline has no reset before delivery or deadline. The revised design delivers at 5.6 ms and resets at 12.1 ms, after completion. Both keep the event latch volatile and have zero pre-delivery resets. A legacy success-and-any-reset label marks the revised design as reset retention, but these traces do not show pending state crossing reset. The checker and legacy labels remain intact; their interpretation follows capture–reset–delivery order.

The external decision is to reject promotion and retain the experimental artifact. Proposal plus revised design costs 14 requests and USD 0.0729432, versus the baseline's five requests and USD 0.0393375. Across the four episodes in this finite batch, including an independent incomplete interface-consumer attempt, all 25 requests settle at USD 0.1766922 upper. Seven recorded new local computational cases include the qualification grid and consumer/design work; the editorial readback adds none. This comparison motivates a prospective protocol with matched cache exposure; it is not retroactively repaired by it.

5 Prospective comparison: a strong baseline saturates the tasks

5.1 Conditions, parity, and frozen sampling

The next protocol compares three method conditions under complete-raw information: fixed H_0, ordinary reflection, and a typed dependency/evidence-scope procedure. Complete-raw supplies all declared sources and the fixed model definition; the host retains full raw traces for audit. Ordinary reflection reviews the provided observations and proposes a method or no update. The typed procedure additionally requests a D/H distinction, source-dependent evidence scope, event ordering, and separation of necessary constraints from sufficient heuristics. It supplies no winning device values. Both proposing conditions must produce actual model-authored hypothesis, falsifier and method text.

Each task/condition/repetition group has the same total cap of 24 requests, 1,200 seconds and two distinct D candidates. The proposal subcap of eight requests and 400 seconds is included, leaving less for fresh design when it is used. Acquisition, failed actions, proposals, no-update decisions, fresh design and confirmation are charged. Equal ceilings do not imply equal realized spending. A no-update retains its cost and uses the base method for an admitted fresh design.

Every arm starts from the same immutable cache manifest of 12 prior responses plus one registered derived baseline. Job overlays are isolated. Later exact-input deduplication does not expose another arm's output before its query or increase the logical two-design allowance. All 28 matched proposal groups, including pilots, pass the recorded information, tool, model-alias, cache and total-budget parity checks. These are eligibility checks, not evidence of benefit.

Two pilot repetitions reverse condition order. The registered rule selects two formal repetitions only if service and design signatures are stable in every matched arm; otherwise it selects three. The pilot signatures are not stable, so three repetitions are frozen once before formal execution. No original failure is replaced and no outcome-dependent extension follows that freeze. The complete eight-condition matrix has 16 pilot and 96 formal groups, comprising 140 episodes and 771 requests. The method subset uses six pilot and 36 formal groups within that matrix. Its fixed-method cell is shared with the interface study and counted once.

5.2 Service ties and a negative retention result

Method	PASS/groups	Requests	Updates	USD upper
Fixed H_0	12/12	64	—	0.4231806
Ordinary reflection	12/12	117	10	0.8190117
Typed procedure	12/12	127	12	0.9367914

Table 1: Fresh method comparison on four fixed digital tasks times three repetitions. All proposal and fresh-design requests are included. The base cell is shared with the interface matrix and counted once.

Each method passes all 12 formal groups, with task-success vector $(3, 3, 3, 3)$. Ordinary reflection proposes ten updates and returns no-update twice; the typed procedure proposes in all twelve. Their total requests are 117 and 127 versus 64 for fixed H_0, including proposal and fresh design. The respective means are 9.750, 10.583 and 5.333 requests per group. All no-update costs remain. Different intermediate reasoning or device choices add no complete-service success to this saturated baseline.

The registered rule requires strictly more task-level service success than both controls, no lost task, and no higher mean request count. The proposed procedure meets neither the strict success improvement nor the request criterion. The selection therefore keeps the unchanged base H. This is a negative result on the registered objective, not a claim that reflection can never help. Four public digital tasks and repeated sampling do not estimate performance over a broad population of products. The H artifacts are instantiated locally per group; no single global H was transported unchanged across these tasks.

The full matrix costs USD 5.4478605 upper: 112 pilot requests at USD 0.7345845 and 659 formal requests at USD 4.7132760. All 771 settle. The 308 requests in the main formal method subset already belong to that total and are not charged again as a separate campaign.

5.3 Correct recomputation without local savings

The response audit annotates 107 existing admitted group responses; five groups have no admitted final response. Of those 107, 91 change D and 16 are unchanged controls. A frozen old trajectory is ineligible after a changed coupled input. Dependency, scope and full-feedback strategies share one actual complete-response execution key in every admitted record. The first two conservatively expand to full response, demonstrating no valid smaller suffix or speed advantage. For a global H change, all valid policies require fresh task-entry decisions. This deterministic audit adds zero scientific execution and is not four new paid strategy arms.

Four formal groups choose a pending event delivered across reset: two fixed method groups, one typed-procedure group and one table-interface group. All reuse the same historical latched-reset response at 10 μF / 45 ms: capture at 3.7–7.9 ms, brownout at 12.1 ms, and reception of the same event at 46.9 ms. Four fresh choices therefore represent one historical ideal response, not four physical experiments or a newly invented memory mechanism. The other 83 successful formal groups deliver before any reset. Physical qualification remains UNKNOWN in every case.

6 A model-authored improver and frozen descendant transfer

6.1 From original experience to G_1

The next study holds the base H_0 unchanged and compares ordinary reflection G_0 with an actual model-authored proposal procedure G_1. One formation episode reads an admitted packet drawn from four retained public method/design episodes. Both proposer conditions receive the same mechanical model-history packet: 75,955 bytes with a recorded hash. The larger 257,661-byte host history is not the model packet. The original-service O0 task and every final input are excluded from formation and selection. The final design entries receive their own current frozen task facts, including the waveform; those facts are not secret from the executing design agent.

The single G_1 formation uses three requests, USD 0.0269157 upper and 38.72 summed seconds. Its 4,599-character procedure asks for a capability ledger, typed symbolic methods, per-world latch/source branching, service-first triage, retrospective rehearsal, an explicit falsifier, cost/error retention and a legal proposal channel. It permits no update. It changes procedural text, not model weights. Eligibility and a source audit do not establish that every generated rule is scientifically valid. In particular, its broad instruction to treat events after the last transmission as benign can fail for ongoing or multiple-event service, and assumed source/channel fields may be unavailable. We retain these defects without hand-editing the generated procedure.

For each of three paired lineages, G_0 and G_1 independently generate two H siblings starting from H_0, giving twelve actual proposals. There is one shared G_1 formation, not three independent training replications. Every proposal is scope-eligible, with scientific validity not established at submission. None is repaired or excluded by hand. Six G_0 proposals use 20 requests and USD 0.1832424; six G_1 proposals use 23 and USD 0.2658975. The combined formation/proposal stage costs 46 requests and USD 0.4760556.

Fifteen recorded unknown-public-source tool failures occur across five proposal episodes, including one G_0 episode and four G_1 episodes. These are source access failures, not unresolved billing or native executions. They remain in traces and costs. They do not justify treating all model descriptions of "schema rejection" as confirmed schema failures.

6.2 Selection, host fault, and a bounded amendment

Each sibling is evaluated from fresh task entry on two development tasks, giving 24 descendant selection cells plus six H_0 controls. Selection ranks more service passes first, then fewer wrong service judgments, then fewer proposal-plus-effective-selection requests, then stable child index. USD is reported but is not the tie criterion. The selected consumed instruction is unchanged H_0 followed by the original model-authored method body; its canonical JSON-string hash is distinct from the proposed body hash.

During selection, three original attempts encounter a host trace-cache error and a fourth is censored. The loader finds cached observations without the separately sealed trace CSV. This is an apparatus fault, not method inferiority. The hold snapshot closes paid dispatch while retaining the original attempts. An explicit amendment repairs CSV loading and provenance only; it preserves the original adapter, task/world/evaluator semantics, payloads, G/H texts, model configuration and resource caps. Six saved-data qualification checks restore the historical assessment, preserve five valid selections, and reject missing or misbound trace/input/world variants. They add no model or scientific execution.

Four separately registered recoveries reuse the original payloads and return PASS. They do not regenerate G or H, replace the original failure records, or become automatic resamples. The four original incident attempts

cost 21 requests and USD 0.1318473; recoveries cost 20 requests and USD 0.1182666. There are 73 canonical cells and 77 actual attempts. Formal sibling ranking uses effective cells; full pipeline and global costs include every original incident and recovery.

Lineage	h_1 requests	h_2 requests	Chosen	H hash prefix
L1-G0	14	13	2	`76060e3a5bb3`
L1-G1	15	12	2	`ba3c85390ea1`
L2-G0	13	13	1	`bc7d9af16574`
L2-G1	14	13	2	`206485c17bc5`
L3-G0	15	15	1	`dde0e5465cf7`
L3-G1	14	13	2	`1d206feb1c02`

Table 2: Every sibling passes both effective selection tasks with zero judgment errors. Requests include that sibling's proposal and its two effective selection entries. All sibling costs remain in full costs.

All thirty effective selections pass. Every sibling passes two of two with zero wrong judgments, so requests and stable index determine the six selected children (Table 2). All six choices and their exact text hashes are frozen before any final input is executed. The portable audit checks the ranking, original authored text, unchanged prefix, selected hashes and actual consumed final instructions. Selection is not a favorable final-result filter.

6.3 Final digital transfer and the unexposed waveform

The final task set contains a stretched BF supply-window derivative, a C-Lite half-power query derivative, and an original-service unexposed waveform. The first two are newly frozen within-model parameter cases, not new physical families. Previously used ESR-failure and no-energy controls are historical, not unseen tests. The study runs each selected G_0/G_1 descendant and a fresh H_0 control on all three final tasks per lineage: 27 entries total.

Arm	Cells	PASS	Submitted FAIL	Censored	Wrong judgment
H0	9	7	1	1	1
G0	9	6	1	2	1
G1	9	8	0	1	0

Table 3: Final transfer outcomes. Censored attempts stay in the denominator. Three descendant-lineage pairs are conditional on one shared G_1 formation.

Every arm passes both digital derivatives: 18/18 cells in total. All differences occur on the waveform task. Its H_0 outcomes are one PASS, one submitted FAIL and one censored; G_0 has zero, one and two; G_1 has two, zero and one. This produces the aggregate 7/9, 6/9 and 8/9 counts in Table 3. The two submitted failures incorrectly claim PASS under the nominal service checker. The four censored cells end with length-limited thinking output at 32,768 tokens and no final submission. They retain their cost and denominator position; they are neither submitted service failures nor evidence of an apparatus error.

The successful G_1 waveform designs in lineages 1 and 3 sample every 0.75 s; the failed submitted designs sample every 1 s and miss event timing. The H_0 success in lineage 2 also samples every 0.75 s. This inspection identifies a device-level difference without attributing a unique mechanism to G_1. The nominal independent O service gate covers energy and event matching, not every inherited A/R requirement. It cannot override the original product's strict FIFO, full dark-service or physical qualification failures.

A separate initial O0 transfer includes a candidate, a fixed-periodic control and a rich-state control. Their nominal service results are PASS, FAIL and PASS. The candidate reports UNKNOWN despite its nominal PASS, and therefore has a wrong nominal judgment. It uses eight requests and USD 0.0760389; the fixed control uses five and USD 0.0321168, and the rich control five and USD 0.0497463. The candidate shows no service advantage over rich state. This task remains outside the G/H formation and selection history.

6.4 Full costs and the inferential limit

Full G_0 cost is 135 requests, USD 1.2585825 and 2,373.96 summed seconds. Full G_1 cost is 138 requests, USD 1.2447618 and 2,304.81 seconds, including the single formation, all siblings, selection, incident attempts, recoveries and final entries. The formation is charged once globally and allocated by thirds across the three G_1 arms. Final-only costs of 40 and 45 requests would omit substantial improvement cost. Excluding only the four original host-incident attempts gives 123 / USD 1.1843580 for G_0 and 129 / USD 1.1871390 for G_1; recoveries remain included. H_0 selection plus final uses 76 requests and USD 0.5154765.

The complete stage totals are 18 requests for O0 transfer, three for formation, 43 for twelve proposals, 173 for 34 actual selection attempts, and 130 for final entries: 367 settled requests, USD 3.1767228 and 5,951.64 summed seconds. The study records 25 new digital scientific executions and zero native dispatches. Cached raw responses, logical design evaluations and requests have different denominators. These are source-bound usage estimates, not invoices.

The G_1 final difference is descriptive. Three lineage pairs share one formation, the task set is small, and only one task distinguishes outcomes. H_0 versus descendant comparisons additionally differ in historical-text exposure; the common-history G_0/G_1 comparison is better matched but still conditions on this one formed G_1. No repeated independent formation, stable recursive self-improvement, learned model weights or physical generalization is established. The registered outcome retains unchanged H_0.

7 A/R engineering consequence: a conditional repair still fails

The A/R tag provides a separate continuous device-level case. Its retained reference has 348 samples on a six-hour profile and first fresh reception at 773 s. In a twelve-hour dark history profile it retains all 170 records, including 168 whole-profile temperature samples, yet 648 of 720 historical dark-window bins contain no new sample. All 73 backlog records miss the recovery deadline and one of two required temperature events is missed. In gateway recovery, all 60 queued records are acknowledged within 82 s and duplicate arrivals cause no duplicate acceptance, but one fresh sample waits 30 s. Retention, new sensing, eventual upload and fresh delay are distinct gates.

One actual Pi episode stops at a host SSE response envelope before submitting a device. An explicitly authorized continuation resumes the same session with its eight prior messages and four tool results. Only the response envelope increases from 2 MB to 64 MiB; the original deadline and cumulative 60-request, three-candidate, 3,600-second caps remain. This is one continued episode, not an independent resample. The originally unresolved request later settles, while its original UNKNOWN snapshot and failure remain saved.

The participant evaluates three adaptive D revisions on four public profiles, giving twelve local case simulations. All use $C = 0.015$ F, thresholds 2.3/2.7 V, 60 s sampling, 1 s ACK spacing, and a rank expression $0 - \text{seq}$. The consumer selects the lowest score, making this newest-first. Retry times are 2, 30 and 8 s; the participant selects candidate 002 with 30 s retry. All three pass applicable energy/activity/receiver checks and fresh delay under the rank interface in the gateway case. The selected candidate still fails dark sampling and recovery service. Its 168 samples are whole-profile counts, not samples collected entirely in darkness.

Candidate	Retry (s)	History samples	Late backlog	Missed events	FIFO
001	2	106	10	2	FAIL
002 (selected)	30	168	73	1	FAIL
003	8	129	34	1	FAIL

Table 4: All A/R alternatives remain in the record. Sample counts cover the whole history profile. Conditional fresh-delay improvement does not satisfy the unchanged dark-service and oldest-first requirements.

The frozen task requires the oldest unacknowledged record first. Independent pending-record replay finds ordering violations for every revision in both history and gateway profiles. The selected candidate's first gateway witness occurs at 833 s: record 2 is chosen while record 1 remains the oldest pending record. The

syntactically legal rank hook cannot authorize a semantic change to FIFO. The contract is not rewritten to accept the conditional timing repair.

An external necessary-bound calculation further establishes NO_GO in the fixed single-capacitor, fixed-sleep legal apparatus. Optimistically omitting boot, sensing and transmission costs, startup by 1,800 s requires $C \leq 0.0528099$ F. Even granting full charge at dark entry and 122 s edge slack, 12-hour coverage at only 5 μW sleep needs $C \geq 0.0712033$ F. The disjoint bounds are a fixed ideal-space result, not physical impossibility or a participant discovery. The participant had access to developer-provided necessary inequalities; the final external proof uses the weaker sleep-only argument independently. No sufficient redesigned device or new hardware is implied.

The continued episode's seven cumulative requests cost USD 0.0826050 upper; the original two requests are already included in that cumulative ledger and are counted once. Product status remains NOT_ACCEPTED, dark service FAIL, strict FIFO FAIL and physical qualification UNKNOWN. Its actual candidate JSON, queue program, source contract, seals, case receipts and ordering witness are delivered for inspection. Engineering files for the separately accepted nominal piezo-node reference, an older report candidate, and O transfer are different identities and cannot qualify A/R candidate 002.

8 Accounting, exposure, and developer intervention

Finite batch	Requests	USD upper
Original four H attempts	68	1.1429220
A/R continued episode	7	0.0826050
Posthoc method and E5	25	0.1766922
Eight-condition comparison	771	5.4478605
Transfer and G lineages	367	3.1767228
Total unique finite requests	1238	10.0268025

Table 5: Non-overlapping finite campaign costs. Provider estimates use frozen rates, not invoices. Historical UNKNOWN and unpriced development assistance remain separate.

The finite campaign contains 1,238 unique settled provider requests and USD 10.0268025 at the upper estimate, including the original four-method attempts. Phases containing A/R, posthoc recovery, the complete eight-condition matrix and transfer use 1,170 of these requests and USD 8.8838805. Shared interface cells occur once. The saved ledger independently binds request ID, owner, status, usage, response hash and charge; the portable readback recomputes token charges and joins episode ledgers to this unique request set.

At the frozen rate schedule, the conservative estimate charges all prompt tokens at USD 0.30 per million and completion tokens at USD 1.20. A separate cache-aware estimate uses USD 0.006 for hit prompt tokens and USD 0.30 for misses. Neither is a provider invoice. The 24 historical UNKNOWN rows from other retained ledgers remain individually present, outside the finite request denominator; their zero charge placeholders are not zero-cost claims. Developer/Codex assistance is unpriced and nonzero, so complete end-to-end cost remains UNKNOWN. Censoring, failed tools, unselected proposals, cache construction and confirmation work are not erased by later success.

The provider reports the alias `deepseek-flash`. The frozen later configuration is hash-bound; independently pinned weights/version are UNKNOWN. The initial operator-written prompt is not a model-authored improver. The posthoc adapter is developer intervention, not a successful original submission. Later prospective role separation and cache matching improve eligibility without repairing that earlier history. The cache-loader amendment is a developer repair with original failures retained. Formation, selection and final input exposure are separately indexed; the host's full retained history must not be mistaken for material actually shown to the model.

All reported outcomes are nominal digital results. No hardware power, temperature, RF, endurance or manufacturing acceptance is supplied. The saved 343-test integration run applies to the registered coordinator

CI profile; it is reused as software evidence and was not rerun for this paper. A passing CI, package hash or PDF build cannot substitute for scientific or product qualification.

9 Discussion and related work

The experiment distinguishes evidence validity from improvement. Conservative invalidation can be correct while saving no coupled computation. A recovered method can produce a valid device while failing a fair-comparison gate. A prospective method can pass every task without adding a success to the strong baseline. A formed improver can yield favorable descendant counts without establishing repeated self-improvement. These are different claims, each with its own denominator and falsifier.

Build-system work provides an executable decomposition of dependency and rebuilding policies [1]. That is a strong conceptual comparison for evidence invalidation. Our coupled examples demonstrate a conservative full-response boundary; they do not outperform such systems or establish a general incremental algorithm. GEPA uses reflected execution histories to propose, test and combine prompt updates [2]. Our ordinary reflection control is a simple registered procedure, not a faithful GEPA implementation, and no direct GEPA comparison is claimed. The present evidence supports an audit of a finite apparatus and of its negative results, rather than a broad optimizer ranking.

DSPy expresses LM pipelines through declarative modules and optimizes them against a chosen metric [3]. STOP improves an LM-calling improver program, distinguishing scaffold revisions from changes to LM weights [4]. Thus, optimizing procedures and separating improvement levels predate our $D/H/G$ notation. Our study instead fixes battery-free service semantics and audits fresh decisions, exposure, censoring and complete selection costs. Neither DSPy nor STOP is directly evaluated here. For intermittent execution, Alpaca uses task-based execution and privatized state updates to support safe restart [5]. We do not implement or outperform that memory mechanism: our cross-reset witness is an inherited ideal latch, and retained state alone does not supply missing sensing energy.

The strongest method control already succeeds on every formal task. This ceiling limits sensitivity to an improved procedure. The later waveform task creates variation but simultaneously exposes long reasoning-output censoring. The data cannot identify how much of a descendant difference comes from specific method clauses, text length, stochastic decisions or the chosen formation history. No component ablation, inferential significance test or population-level success-rate estimate is justified by this task count.

The next evidential step for an improvement claim would require independently formed G_1 procedures, prospectively frozen new tasks, matched exposure and retention rules, and a budget covering all attempts. It would also need to separate source acquisition, service judgment and device success. A shorter or cheaper final design alone would remain insufficient. Those experiments are not performed in this delivery. The observed decision is to retain H_0, the original failures, and the limited scope of every computational PASS.

10 Reproducibility and engineering handoff

The accompanying package contains one independent manuscript and a finite saved-data reproducer. Its Python core uses the standard library, accepts `-data-root` and `-output`, and never imports the apparatus, opens an embedded historical host path, calls a model or reruns a response. It checks every manifest-relative input hash, recomputes statistical tables, replays sibling selection from saved metrics, binds actual selected method text to final consumption, and joins usage rows without double-counting the continued A/R episode. Missing, modified and foreign-candidate input controls must be rejected. Historical absolute artifact locators are retained as inert provenance in original JSON, not runtime dependencies.

F1 reconstructs response-validity controls; F4 reconstructs early and prospective method outcomes; F5 reconstructs selection, transfer, censoring and complete costs; F6 checks the saved failed A/R baseline. F2 and F3 explicitly report no decision-span ablation or organization comparison. The package does not turn these unperformed experiments into results. A data dictionary, claim–evidence index, model/exposure and intervention index, exact source list and candidate-bound engineering handoff accompany the code. The paper's tables are generated from those inputs. A local two-pass TeX build and individual-page visual review

qualify the delivered PDF; coordinator acceptance in a detached directory is a separate recorded step, not assumed here.

References

[1] Andrey Mokhov, Neil Mitchell, and Simon Peyton Jones. Build systems à la carte. ICFP, 2018. https://www.microsoft.com/en-us/research/publication/build-systems-la-carte/.

[2] Lakshya A. Agrawal et al. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. arXiv:2507.19457v2, 2026. https://arxiv.org/abs/2507.19457v2.

[3] Omar Khattab et al. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. arXiv:2310.03714, 2023. https://arxiv.org/abs/2310.03714.

[4] Eric Zelikman, Eliana Lorch, Lester Mackey, and Adam Tauman Kalai. Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation. COLM, 2024. https://arxiv.org/abs/2310.02304v3.

[5] Kiwan Maeng, Alexei Colin, and Brandon Lucia. Alpaca: Intermittent Execution without Checkpoints. Proceedings of the ACM on Programming Languages 1(OOPSLA), Article 96, 2017. https://abstract.ece.cmu.edu/pubs/alpaca.pdf.

A Saved-data keys and claim boundaries

The portable manifest lists original byte identities; the coordinator's collection receipt additionally gives the exact source filenames. Raw file SHA-256 and canonical JSON content hashes have distinct meanings. Method and candidate content hashes use UTF-8 JSON with sorted keys, compact separators, unescaped Unicode and no nonfinite values. A raw UTF-8 method-body digest is not that canonical string digest. The selected instruction includes the common base prefix; it must not be equated with the proposed body alone.

The four digital method tasks are registered development objects. The stretched-window and half-power cases are within-model digital derivatives. The final waveform is unexposed to formation/selection but fully supplied to the executing design entry. The original-service O0 control is separately excluded from formation. Cross-reset groups share a historical ideal latch response. These exposure classes prevent an aggregate "unseen performance" claim across incompatible histories.

The complete paper package preserves the A/R selected candidate's content identity and both rejected alternatives. It does not attach native engineering assets from a different candidate to supply missing A/R qualification. The engineering handoff records requirements, legal changes, known failures, unknown physical properties and the original ordering witness. Any future service relaxation, new energy source or changed apparatus requires a new contract and qualification rather than relabeling this baseline.